The data class

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Fork and clone repository
https://git.aspp.school/ASPP/2024-heraklion-data.git

Things one thinks about when thinking about data

Storage	Processing	Reproducibility and collaboration
• Size • Access ease • Access time	• Efficient processing • Organizing data so that analyses are easy	• Versioning • Lineage tracing (which script / other data was used to generate this?) • Ease of sharing

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Storage	Processing	Reproducibility and collaboration
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Hands-on

What data structure would you use to represent…

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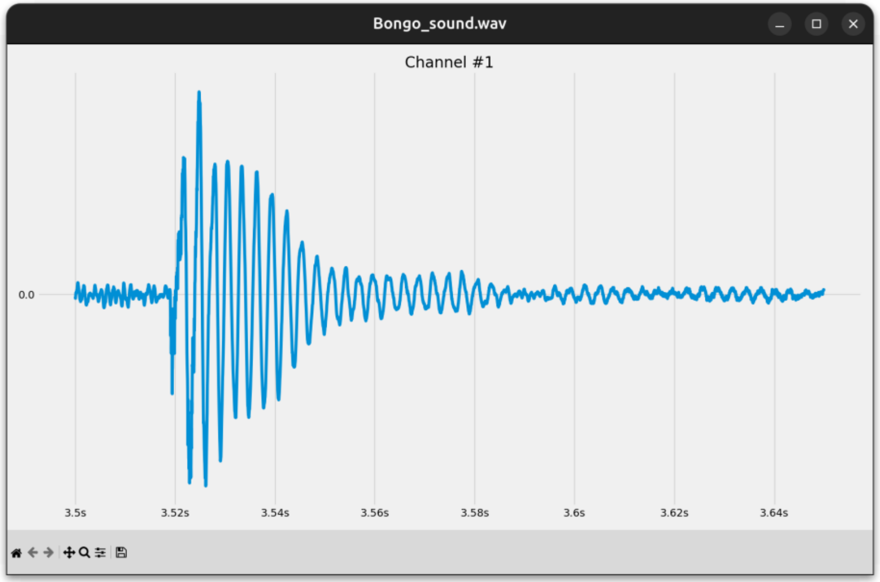
What data structure would you use to represent...

A sound wave?



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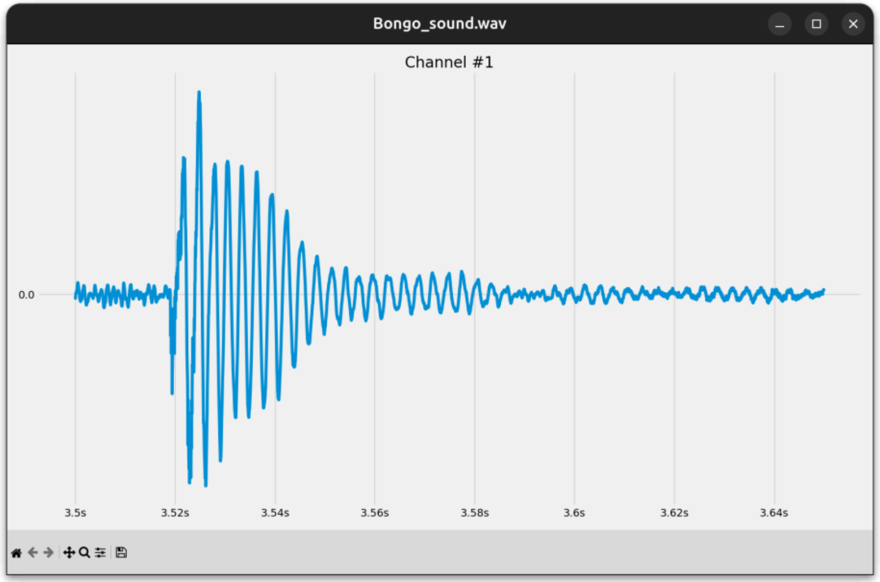


NumPy array

```
In [6]: sound_data

Out[6]: array([0.66709183, 0.55973494, 0.95416669, 0.60810949, 0.05188879,
               0.58619063, 0.25555136, 0.72451477, 0.2646681 , 0.08694215,
               0.75592186, 0.67261696, 0.62847452, 0.06232598, 0.20549438,
               0.11718457, 0.25184725, 0.48625729, 0.8103058 , 0.18100915,
               0.81113341, 0.62055231, 0.9046905 , 0.56664205, 0.73235338,
               0.74382869, 0.64856368, 0.80644398, 0.46199345, 0.78516632,
               0.91298397, 0.48290914, 0.20847714, 0.99162659, 0.26374781,
               0.3602381 , 0.07173351, 0.8584085 , 0.32248766, 0.39167573,
               0.67944923, 0.00930429, 0.21714217, 0.58810089, 0.17668711,
               0.57444803, 0.25760187, 0.43785728, 0.39119371, 0.68268063,
               0.95954499, 0.45934239, 0.03616905, 0.23896063, 0.61872801,
               0.76332531, 0.96272817, 0.57169277, 0.50225193, 0.01361629,
               0.15357459, 0.8057233 , 0.0642748 , 0.95013941, 0.38712684,
               0.97231498, 0.20261775, 0.74184693, 0.26629893, 0.84672705,
               0.67662718, 0.96055977, 0.64942314, 0.66487937, 0.86867536,
               0.40815661, 0.1139344 , 0.95638066, 0.87436447, 0.18407227,
               0.64457074, 0.19233097, 0.24012179, 0.90399279, 0.39093908,
               0.26389161, 0.97537645, 0.14209784, 0.75261696, 0.10078122,
               0.87468408, 0.77990102, 0.92983283, 0.45841805, 0.61470669,
               0.87939755, 0.09266009, 0.41177209, 0.46973971, 0.43152144])
```

Hands-on

What data structure would you use to represent…

A map between color names and RGB values?

Hands-on

What data structure would you use to represent...

A map between color names and RGB values?

Dictionary

```
colors_hex = {
    "tawny orange": "#CD5700",
    "very peri": "#6667AB",
    "iced coffee": "#C5A582”,
    "pink flambé": "#DC4C8B”,
    # …
}
```

Hands-on

What data structure would you use to represent…

Phone book entries?

Hands-on

What data structure would you use to represent…

Phone book entries?

argrave Fiduciary Advisors LLC
3030 Bridgeway Sau 415 729-9283
ARGREAVES David 276 Devon Dr S R 415 448-5180
David & Becky 276 Devon Dr S R 415 479-3016
Gordon 965 Magnolia Av Lrkspr 415 924-2582
S 415 464-0822
William 415 388-3439
William 415 388-4705
ARIRI Farhad & Mojgan 415 332-0287
Farnoosh 187 Cazneau Ave Sau 415 332-7533
ARKAVY Kamila 415 454-3136
Kamila 415 454-3416
ARKER Howard 30 Ralston Av M Vly 415 383-9458
ARKEY Teall 296 Union St S R 415 456-4818
ARKIN John 20 Minor Ct S R 415 472-2452
ARKINS Edward 206 Evergreen Dr Kntfld 415 461-4116
ARLAN Carol R 415 669-7850
David 415 888-2112
ARLAND C 415 663-9283
ARLE Jonathan Gabrielle 415 889-5334

Timothy
Uta
Harrington's Moving & Storage
4415 Paradise Dr Tibrn
HARRIS Adam 106 Baltimore Ave C M
Alan & Christine
Andrew & Mary 8 Via Capistrano Tibrn
Anne 102 Ryan Av M Vly
Anne 102 Ryan Av M Vly
Anne 102 Ryan Av M Vly
Arlene L
B
Harris Bail bonds 775 E Blithedale AV M Vly
HARRIS Barbara
Barbra
Barry
Bernard & Bette
Bernice
Bourke
Brent & Nanette 50 La Cuesta Lagntas

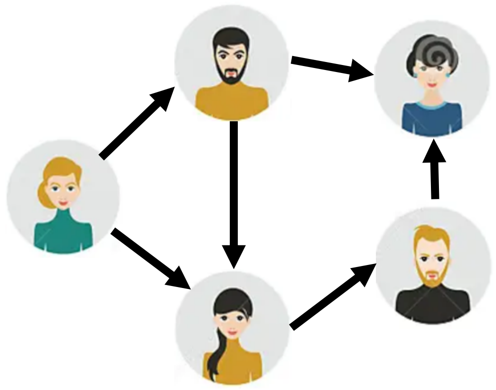
Pandas DataFrame

first_name	last_name	phone_nr	address	ZIP	city
John	Doe	555-1234	123 Maple St	12345	Springfield
Jane	Smith	555-5678	456 Oak St	67890	Rivertown
Alice	Johnson	555-8765	789 Pine St	54321	Lakeside
Bob	Brown	555-4321	321 Birch St	09876	Hilltop
Emma	Davis	555-7890	654 Elm St	11223	Greendale

Hands-on

What data structure would you use to represent...

Friendship relations?



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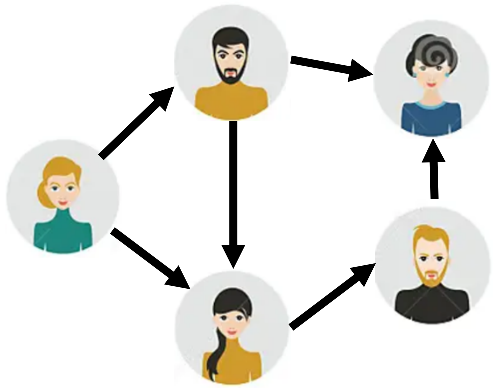
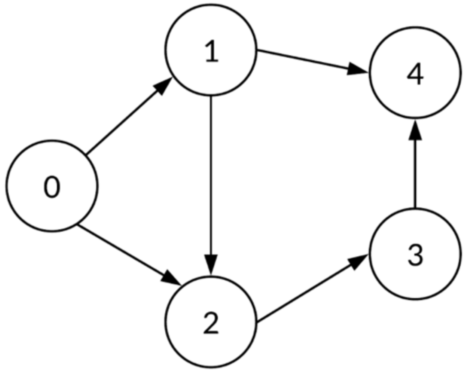
Friendship relations?



Graph



Implemented as

	0	1	2	3	4
0	0	1	1	0	0
1	0	0	1	0	1
2	0	0	0	1	0
3	0	0	0	0	1
4	0	0	0	0	0

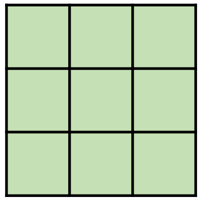
Adjacency matrix (array)

```
A_dict = {
    '0':[1,2],
    '1':[2],
    '2':[3],
    '3':[4],
    '4':[]
}
```

Dictionary

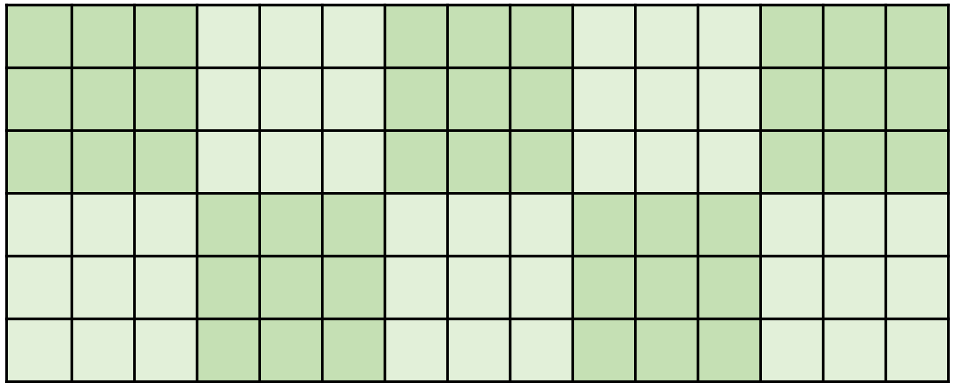
You develop your code on a small data set, how is it going to scale to the complete data set?

Development data



N data points,
Processing time T

Real data

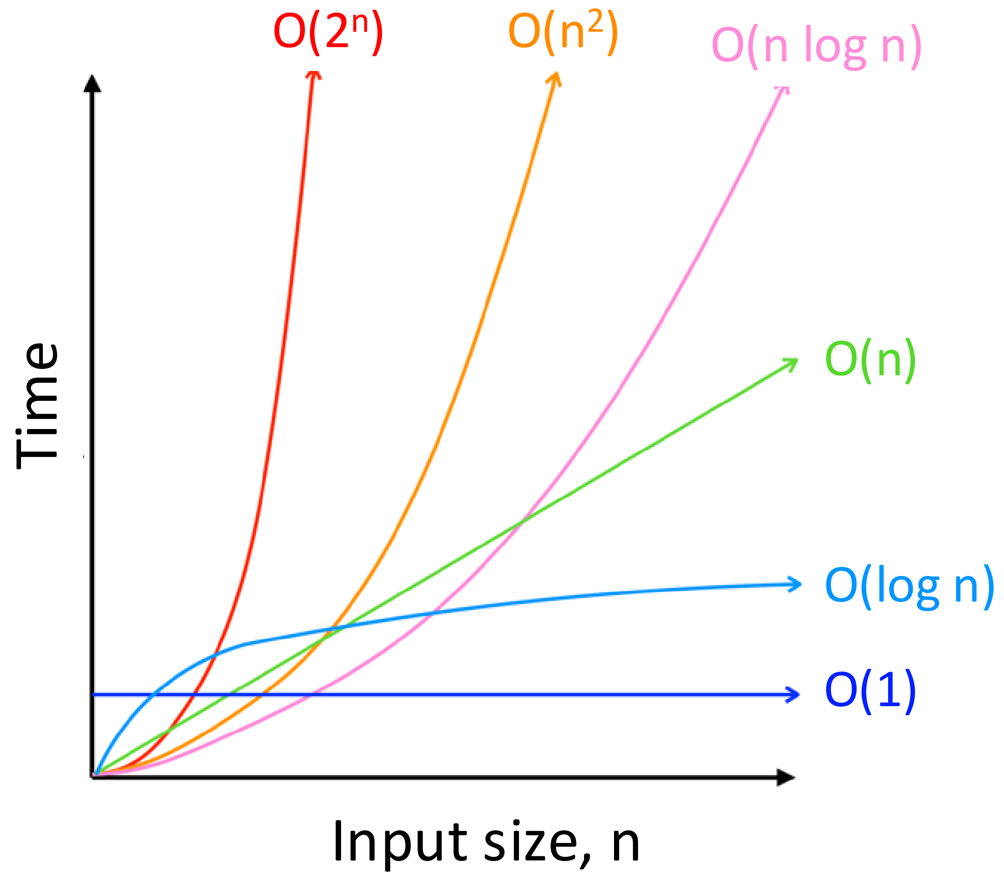


10x N data points
Processing time -> ?

We’re interested in orders of magnitude

How performance scales: big-O

Big-O class	What we call it	Time increase, when data increases 10x
O(1)	constant	1x time
O(n)	linear	10x time
$O(n^2)$	quadratic	100x time
O(n * log n)	linearithmic	~10-20x time
O(log n)	logarithmic	~1-2x time



Hands-on: Operations on lists

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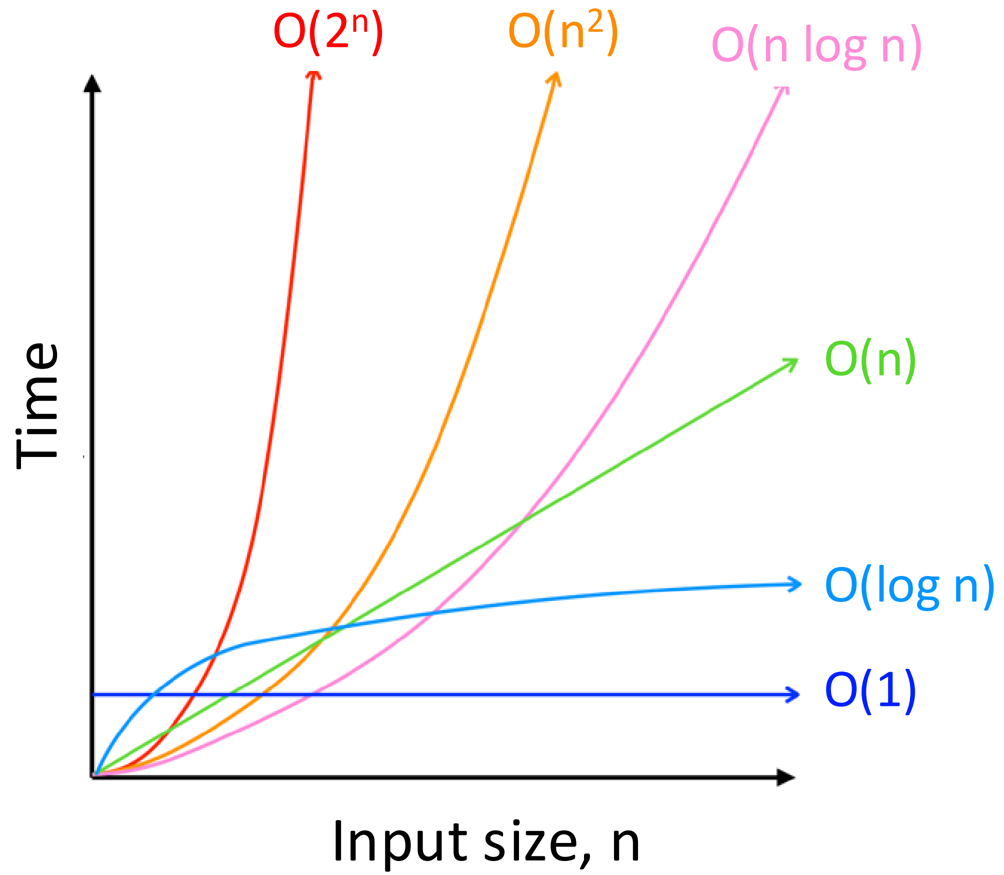
Big-O class	Operation on lists that scales this way
$O(1)$	
$O(n)$	
$O(n^2)$	
$O(n * \log n)$	
$O(\log n)$	



Hands-on: Operations on lists

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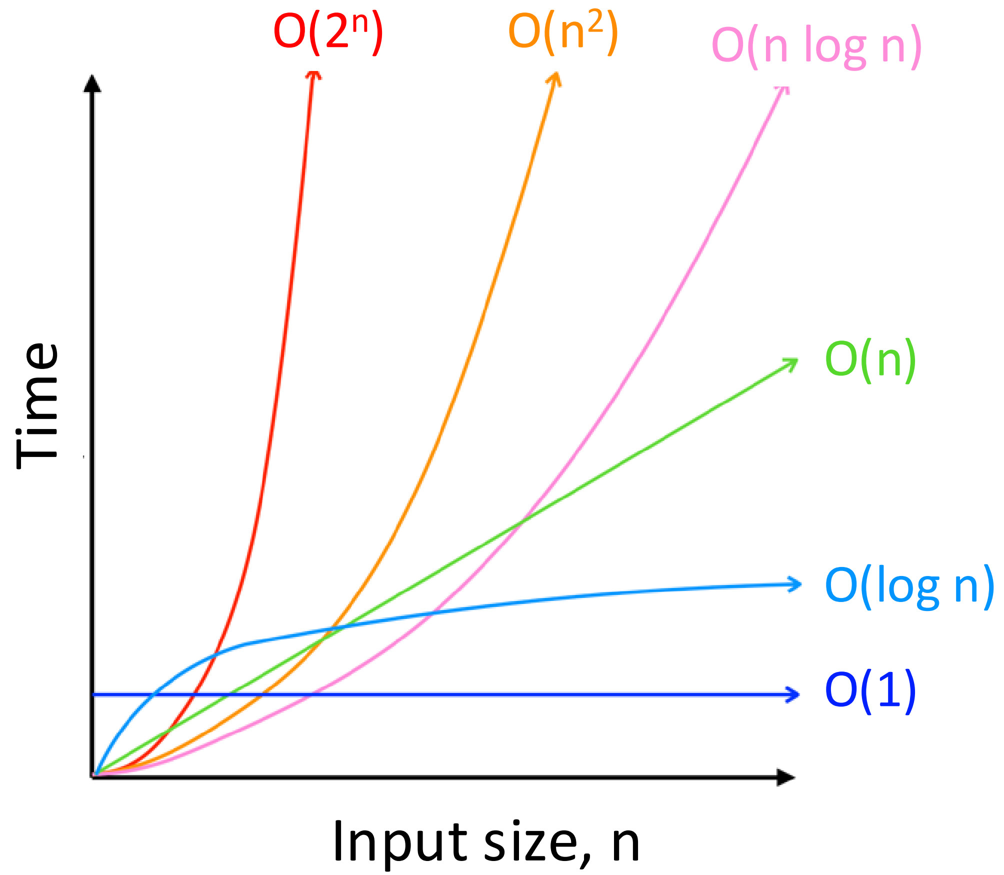
Big-O class	Operation on lists that scales this way
O(1)	Getting an element by its index
O(n)	Summing elements in list
$O(n^2)$	Computing distance between all pairs of elements in the list
O(n * log n)	Sorting the list
O(log n)	Searching an element in a sorted list



Example: Find common words

Given two lists of words, extract all the words that are in common

```
words1 = ['apple', 'orange', 'banana', 'melon', 'peach']
words2 = ['orange', 'kiwi', 'avocado', 'apple', 'banana']
```

Expected result: `['apple', 'orange', 'banana']`

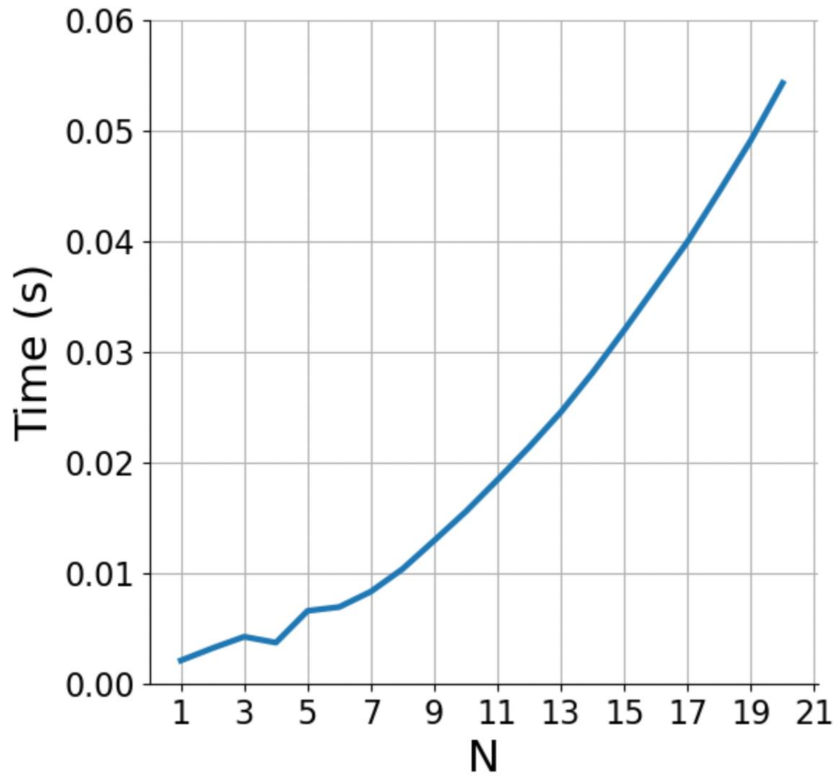
Implementation with two for-loops

```
words1 = ['apple', 'orange', 'banana', 'melon', 'peach']
words2 = ['orange', 'kiwi', 'avocado', 'apple', 'banana']

common = []
for w in words1:
    if w in words2:
        common.append(w)
```

What is the big-O complexity of this implementation?

Implementation with two for-loops

```python
words1 = ['apple', 'orange', 'banana', 'melon', 'peach']
words2 = ['orange', 'kiwi', 'avocado', 'apple', 'banana']

common = []
for w in words1:          # O(N)
    if w in words2:          # O(N)
        common.append(w)  # O(1)
```



What is the big-O complexity of this implementation?

N * N ~ **$O(N^2)$**

Implementation with sorted lists

```
words1 = ['apple', 'orange', 'banana', 'melon', 'peach']
words2 = ['orange', 'kiwi', 'avocado', 'apple', 'banana']

words1 = sorted(words1) # ['apple', 'banana', 'melon', 'orange', 'peach']
words2 = sorted(words2) # ['apple', 'avocado', 'banana', 'kiwi', 'orange']

common = []
idx2 = 0
for w in words1:
    while idx2 < len(words2) and words2[idx2] < w:
        idx2 += 1

    if idx2 >= len(words2):
        break

    if words2[idx2] == w:
        common.append(w)
```

What is the big-O complexity of this implementation?

Implementation with sorted lists

```python
words1 = ['apple', 'orange', 'banana', 'melon', 'peach']
words2 = ['orange', 'kiwi', 'avocado', 'apple', 'banana']

words1 = sorted(words1)     # O(N * log(N))
words2 = sorted(words2)     # O(N * log(N))

common = []
idx2 = 0
for w in words1:                                        # O(N)
    while idx2 < len(words2) and words2[idx2] < w:      # O(N) in total
        idx2 += 1

    if idx2 >= len(words2): # O(1)
        break

    if words2[idx2] == w:   # O(1)
        common.append(w)
```

What is the big-O complexity of this implementation?

2 * (N * log(N)) + 2 * N ~ **O(N log N)**

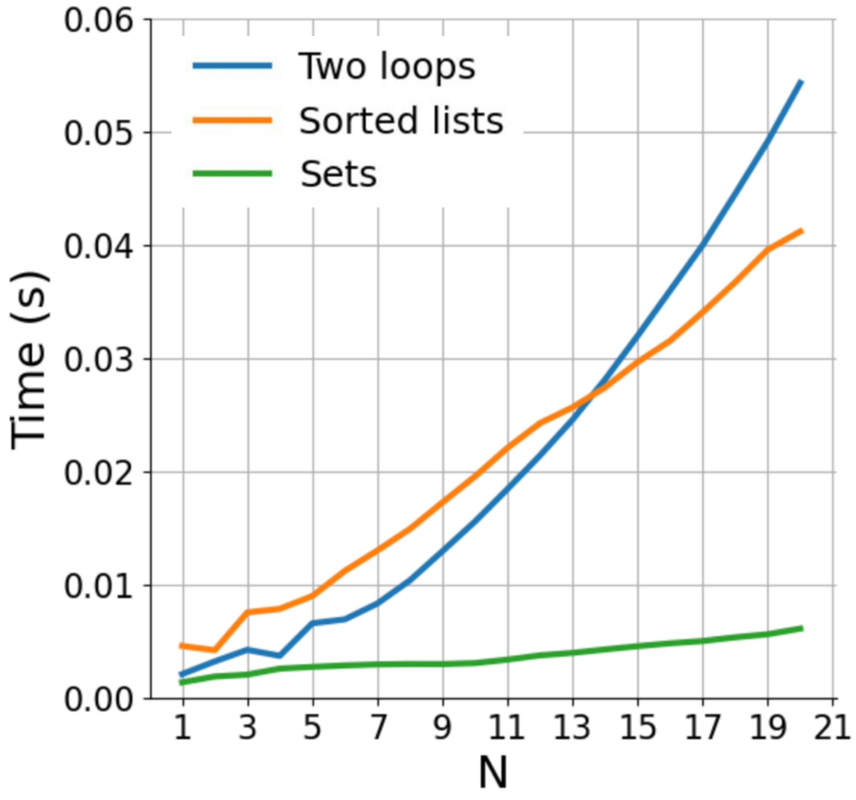
Implementation with sets

```
words1 = ['apple', 'orange', 'banana', 'melon', 'peach']
words2 = ['orange', 'kiwi', 'avocado', 'apple', 'banana']

words2 = set(words2)

common = []
for w in words1:
    if w in words2:
        common.append(w)
```

What is the big-O complexity of this implementation?

Implementation with sets

```python
words1 = ['apple', 'orange', 'banana', 'melon', 'peach']
words2 = ['orange', 'kiwi', 'avocado', 'apple', 'banana']

words2 = set(words2)       # O(N)

common = []
for w in words1:           # O(N)
    if w in words2:            # O(1)
        common.append(w)       # O(1)
```



What is the big-O complexity of this implementation?

N + N ~ **O(N)**

Basic reference sheet about Python data structures

Lists: collection of ordered, arbitrary data

Getting an element by index	O(1)
Appending	O(1)
Inserting an element at index	O(n)
Sorting	O(n log n)
Finding an element (e.g., "if element in my_list: …")	O(n)

Dictionaries ("hashmaps")

Inserting	O(1)
Finding a value by key (e.g., "if element in my_dict: …")	O(1)

Sets: it's dictionaries without values

Inserting	O(1)
Finding a value by key (e.g., "if element in my_set: …")	O(1)

See also: https://wiki.python.org/moin/TimeComplexity

Hands-on

Exercise

`exercises/match_tarots`

- Open the notebook `match_tarots`, and follow the instructions!
- Submit a PR for Issue #1 on `git.aspp.school/ASPP/2026-latam-data`

How does rewriting in C change the performance?

(rewriting in C, parallelization; same algorithm)

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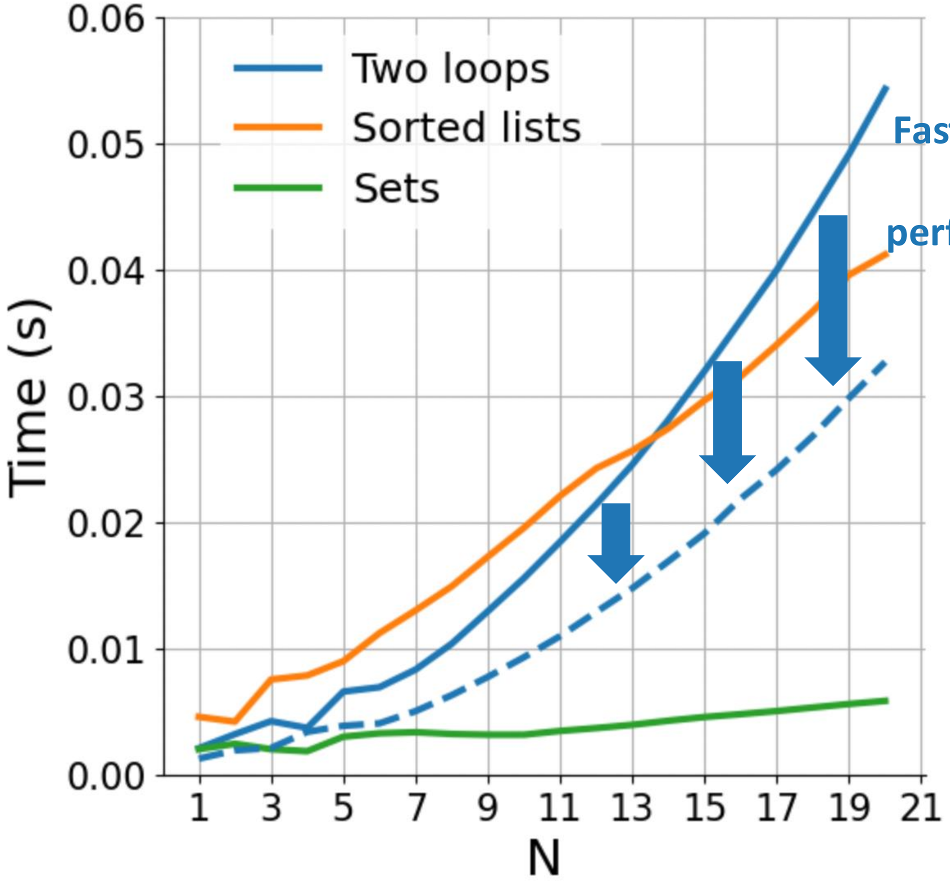
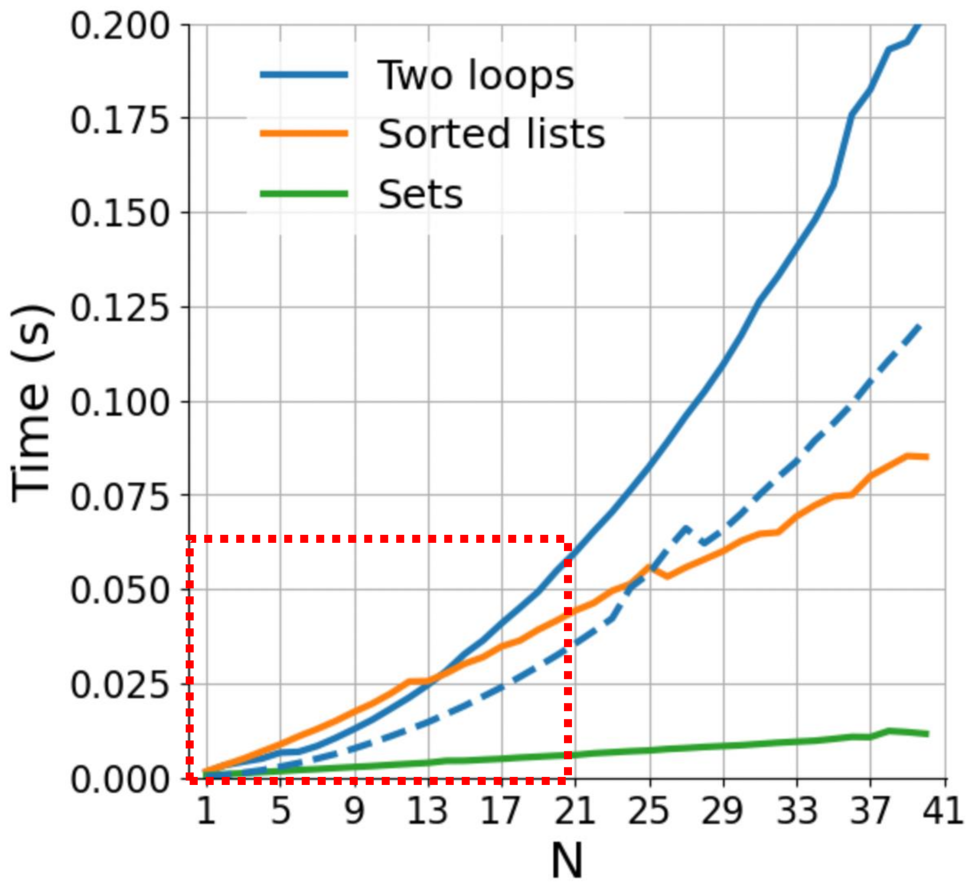


Faster language increases performance for fixed N

How does rewriting in C change the performance?
(rewriting in C, parallelization; same algorithm)



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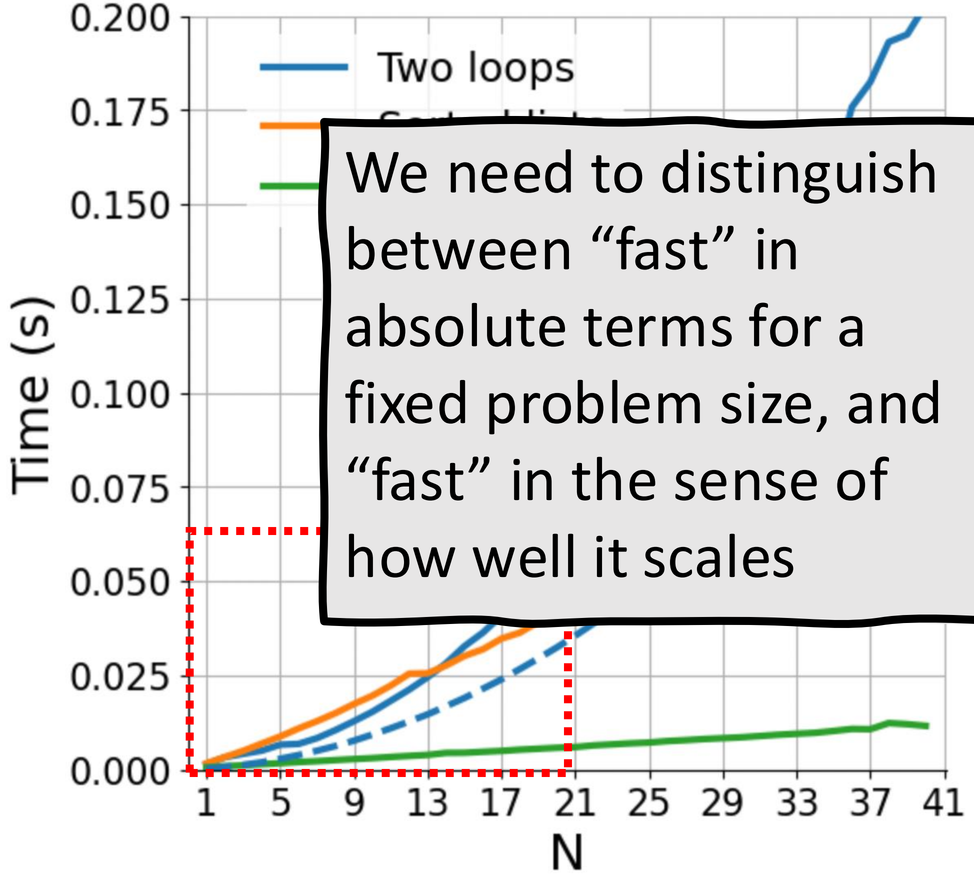

We need to distinguish between “fast” in absolute terms for a fixed problem size, and “fast” in the sense of how well it scales

COMING UP NEXT:
NumPy and the array data structure

Some data structures

- list: ordered, heterogeneous storage, stack/queue, fast access by index, slow search
- set: unordered
- dictionary
- arrays: e.g. numpy, HDF5
- tables: e.g. pandas, dask, spark, SQL
- graph: social network structure
- tree: to rapidly search a dataset
- heap
- stack
- queue
- priority queue